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Group 16

Lab Answers 1: Markov Chains - Smart Home Energy Management

Question 2

2.1. To test if a transition matrix is incorrect, we can verify its fundamental properties such as if

- The sum of the probabilities in every single row are equal to **1.0**. This ensures that from any given state, the system always transitions somewhere;
- Every element in the matrix must be greater than or equal to 0. Negative probabilities are mathematically invalid;
- Because State 75 is a permanent failure state, its row must have a probability of 1.0 to transition to itself (index [75, 75]) and 0.0 for transitioning to any other state.

2.2 To compare the theoretical and empirical probabilities of failure over time $t \in [0, 40]$, we need to compare both the:

- **Theoretical Probability** $\mu_t[75]$: We calculated it using matrix multiplication. Starting with an initial state distribution vector μ_0 (where the initial state s_0 has a probability of 1.0, and all other states are 0.0), the distribution at any time t is found by repeatedly multiplying by the transition matrix P: $\mu_t = \mu_{t-1}P$. The theoretical probability of failure at time step t is simply the value at the 75th index of the resulting vector μ_t. $\mu_t[75]$:
- **Empirical Probability**: We implemented a function to generate N_r independent trajectories, each lasting N_s steps, starting from s_0. At each step, the next state was chosen randomly based on the transition probabilities of the current state. The empirical probability of failure at time step t is the fraction of these N_r simulations that are in State 75 at that specific time step.

We obtained the following results for each one of the 3 scenarios:

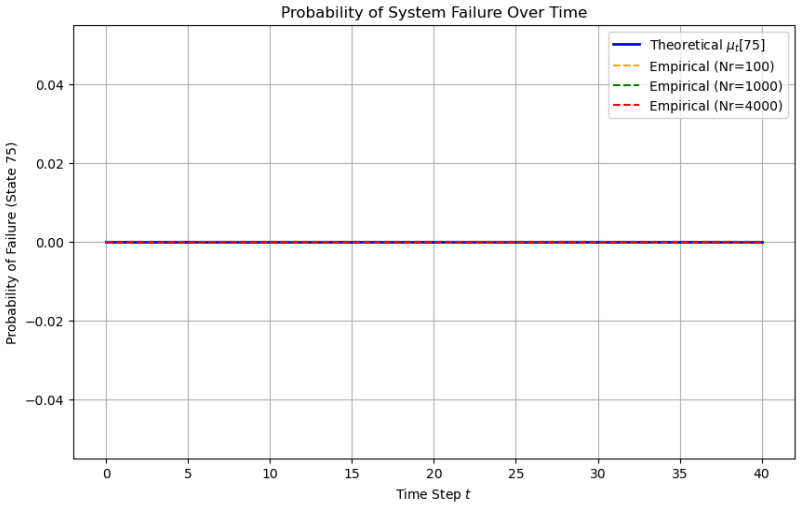

Fig. 1 - Scenario A

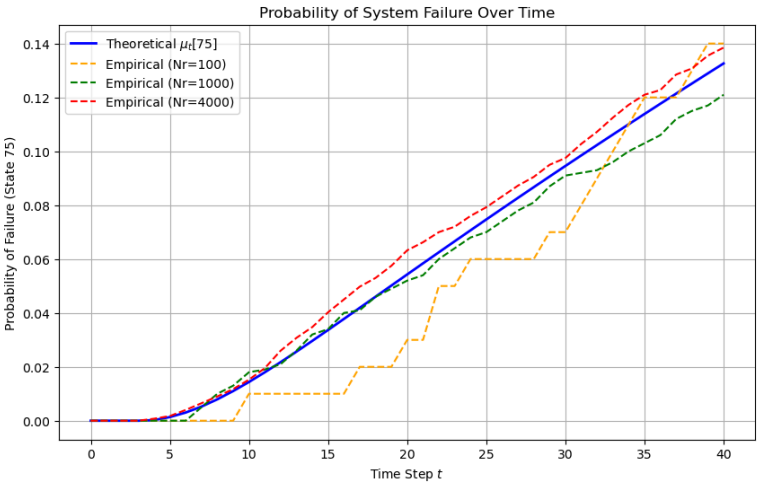

Fig. 2 - Scenario B

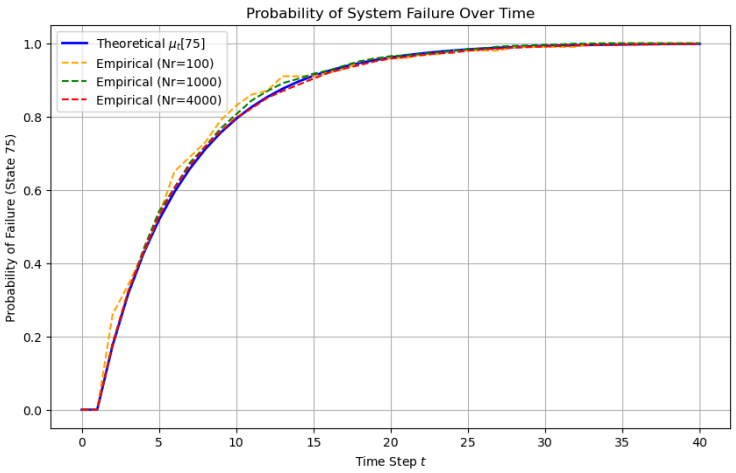

Fig. 3 - Scenario C

The plots successfully validate our models, demonstrating that as empirical simulations increase ($N_r = 4000$), they converge with the mathematical expectation $\mu_t[75]$. As expected, Scenario A remains entirely stable with a 0% failure rate because there are no transitions to the failure state. Scenario B shows a gradual degradation to roughly 13% failure by step 40, reflecting the system's vulnerability when hitting an empty battery. In contrast, Scenario C exhibits rapid system collapse, approaching 100% failure by step 30 due to the increased probability of failure when the battery is either full or empty.

2.3 To compute the stationary distribution of a Markov chain we can use one of the following 3 methods:

- **Method 1 (Eigen-decomposition):** Find π such that $\pi P = \pi$, which basically calculates the left eigenvector of P for an eigenvalue of 1.
- **Method 2 (Power Iteration):** Repeatedly multiply any initial distribution until convergence: $\mu_{n+1} = \mu_n \cdot P \quad \text{until} \quad \|\mu_{n+1} - \mu_n\| < \varepsilon$
- **Method 3 (Linear System):** Solve $\pi P = \pi$ with the constraint $\sum_i \pi_i = 1$ directly via the system $\pi(P - I) = \mathbf{0}$.

The results obtained with these 3 methods depend on the properties of the Markov chain. If the Markov chain is finite, irreducible (all states are intercommunicating), and aperiodic (no deterministic cycles)—meaning it is an ergodic chain—all three methods will yield the exact same, unique stationary distribution. If the chain is periodic, Methods 1 and 3 can still find the algebraic stationary distribution, but Method 2 will fail to converge.

2.4 In Scenario C, State 75 acts as an absorbing sink state. Because there is a constant, non-zero probability of transitioning to this failure state from the boundaries (battery empty or full), and all states can eventually reach these boundaries, the system is guaranteed to fail given infinite time. Therefore, computing the stationary distribution is mathematically trivial: it will always result in a 100% probability of being in State 75 and 0% for all operational states.

Knowing that the system will eventually fail does not help us evaluate its practical performance. Instead, the **Expected Time to Failure** is a much more valuable metric because it quantifies *how long* the system can operate before failing. This allows a homeowner to understand the system's practical lifespan and optimize their initial conditions to maximize operational time.

Finally, for an initial state of 37 (like the one in the previous question) we obtained a expected time of failure at around 7.18 time steps.

2.5

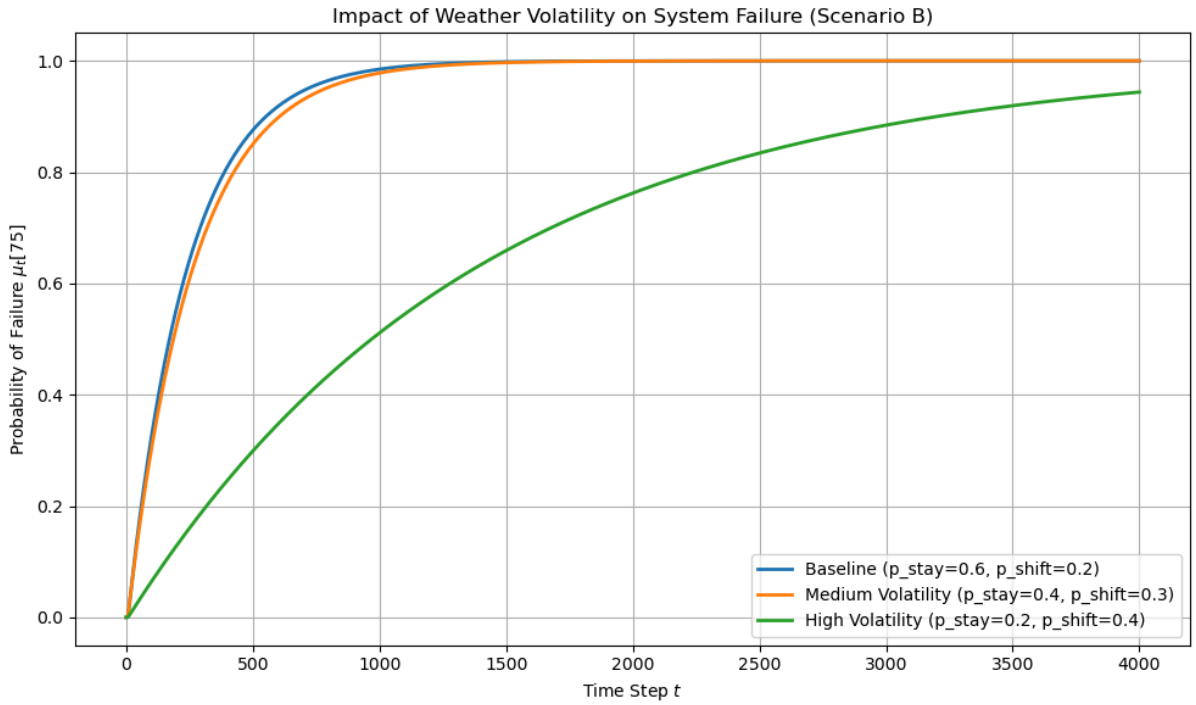

Fig. 4 - Probability of Failure, according to the volatility of the weather parameter (W)

Based on the empirical simulation over 4000 iterations (Fig. 4), increasing the volatility of the Weather Markov Chain (lowering the probability of staying in the same state) decreases the rate at which the system fails in Scenario B. The Baseline system ($p_{stay} = 0.6$) reaches a near 100% failure rate much faster than the High Volatility system ($p_{stay} = 0.2$).

This happens because the system only accumulates failure probability when transitioning to an empty battery ($B' = 0$). The battery updates according to the rule $B' = clip(B + W - C, 0, 4)$. To fully drain the battery from a healthy state to 0, the system must experience a sustained sequence where consumption exceeds generation (W needs to be 0 or 1 for a lot of iterations).

With a 60% chance, if the weather turns to State 0 (Stormy), it is highly likely to remain stormy for several consecutive time steps. This sustained deficit rapidly drains the battery to 0, triggering the failure condition.

With a High Volatility (20% chance), the environment is highly chaotic and changes rapidly. Even if the system enters a Stormy state ($W = 0$), the low probability of staying means the weather is very likely to improve in the next step or two, hence the battery has enough time to fully deplete.

In this specific Smart Home model, a highly volatile environment acts as a protective mean-reverting mechanism. It prevents the prolonged, uninterrupted streaks of bad weather that are necessary to completely drain the battery.

2.6 The homeowner should aim for a battery level of 2 (**B=2**) if they want to avoid State 75 in Scenario C because the system redirects 5% of the transition probability to failure for every next state where **B'=0** or **B'=4**. A homeowner starting at **B=4** is at high risk because:

- **B=4** easily transitions into **B'=4**, which carries an immediate failure risk.
- From **B=4**, with low consumption (**C**), the battery clamps to 4 for any excess generation (**W**). This means many transitions stay at **B'=4**, repeatedly triggering the 5% penalty

Looking directly at the transition matrix for neutral weather (**W=2**) and moderate consumption (**C=1**), we can see the immediate risk. The row for a starting state of **B=4** has a direct transition probability to the failure state (State 75) of 0.25 (or 25%). In stark contrast, the transition matrix row for a starting state of **B=2** under those exact same conditions has a direct transition probability to failure of 0.00 (0%).

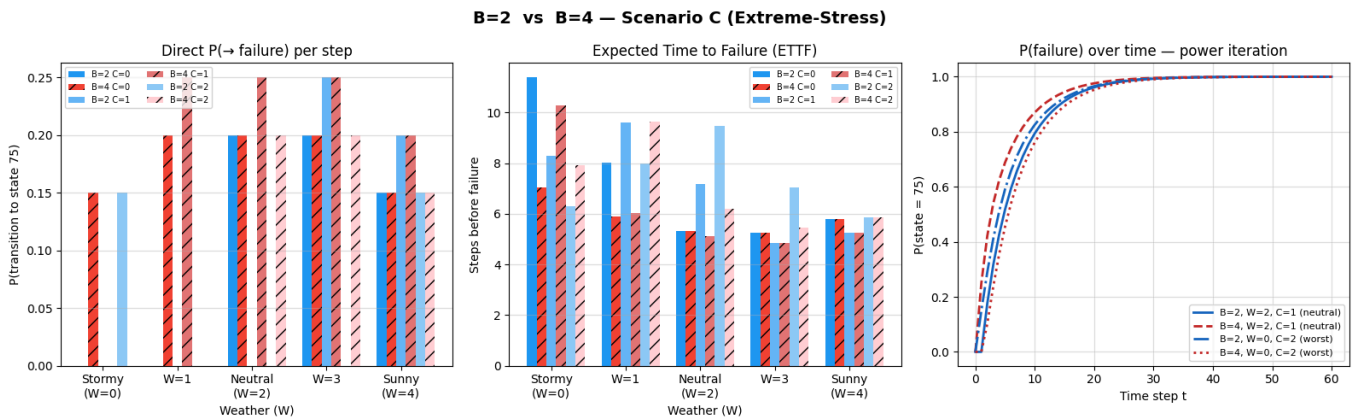

Fig. 5 - Comparing B=2 and B=4

By analyzing the charts, we obtained the following values:

ETTF summary (W=2, C=1 — neutral conditions):

- **B=2:** 7.18 steps
- **B=4:** 5.13 steps

Average ETTF across all (W,C):

- **B=2:** 7.17 steps
- **B=4:** 6.40 steps

As we can see, with a battery level of 2 (**B=2**), we need more steps to reach failure on average, proving it is the safer strategy.